as and when it is required.

Dr Ravinder Dahiya from University of Glasgow's School of Engineering adds, "The next step for us is to further develop the power-generation technology that underpins this research and use it to power the motors that drive the prosthetic hand itself. This could allow the creation of an entirely energy-autonomous prosthetic limb."

Graphene based electrode that increases solar power storage

Researchers at RMIT University, Australia, have developed a groundbreaking prototype that could be the answer to the storage challenge that still holds solar back as a total energy solution. The new type of electrode could boost the capacity of existing integrable storage technologies by 3000 per cent.

The electrode prototype can be combined with a solar cell for on-chip energy harvesting and storage (left); a western swordfern leaf magnified 400 times, showing the self-repeating fractal pattern of its veins (right) (Image courtesy: www.rmit.edu.au)

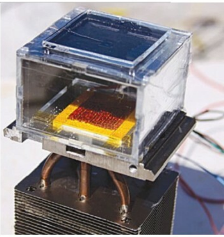
The graphene based prototype also opens a new path to the development of flexible thin-film all-in-one solar capture and storage, bringing us one step closer to self-powering smartphones, laptops, cars and buildings.

The new electrode is designed to work with supercapacitors, which can charge and discharge power much faster than conventional batteries. Supercapacitors have been combined with solar, but their wider use as a storage solution is restricted because of their limited capacity.

Min Gu, professor at RMIT, has said that the new design draws on nature's own genius solution to the challenge of filling a space in the most efficient way possible—through intricate self-repeating patterns known as fractals.

Solar-powered device that can pull water from dry air

Over the years, researchers have developed ways to grab a few trickles, such as using fine nets to wick water from fog banks, or power-hungry dehumidifiers to condense water out of the air. But both approaches require either very humid air or far too much electricity to be broadly useful.

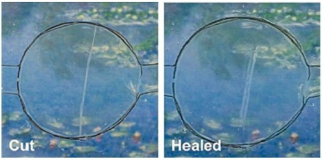


The new water harvester is made of metal organic framework crystals pressed into a thin sheet of copper metal and placed between a solar absorber (above) and a condenser plate (below) (Image courtesy: www.phys.org)

To find an all-purpose solution, a team of researchers led by Omar Yaghi, a chemist at University of California, Berkeley, USA, turned to a family of crystalline powders called metal organic frameworks, or MOFs. Yaghi developed the first MOFs—porous crystals that form continuous 3D networks—more than 20 years ago. The networks assemble in a Tinkertoy-like fashion from metal atoms that act as hubs and stick-like organic compounds that link the hubs together. By choosing different metals and organics, chemists can dial in the properties of each MOF, controlling what gases bind to these, and how strongly these hold on.

Scientists create self-repairing smartphone screen material

Scientists, including several from University of California, Riverside, USA, have developed a transparent, self-healing, highly-stretchable conductive material that can be electri-



The polymer invented by Chao Wang is the first of its kind in terms of electrical conductivity (Image courtesy: (www.sciencedaily.com)

cally activated to power artificial muscles and could be used to improve batteries, electronic devices and robots. The project brings together the research areas of self-healing materials and ionic conductors.

Inspired by wound healing in nature, self-healing materials repair damage caused by wear and tear, and extend the lifetime and lower the cost of materials and devices. Chao Wang, an adjunct assistant professor of